

SRE INTERVIEW PREP: LINUX INTERNALS & NETWORKING

Target Roles: Senior SRE, Senior DevOps, Lead Platform Engineer

Target Companies: Google, Apple, Nvidia, Stripe

Reference Material: *The Linux Programming Interface (TLPI)* by Michael Kerrisk

GROUND RULES

- **Infrastructure-First:** Focus on how the kernel manages resources, not how to write application code.
- **The "Why" Matters:** For every concept, ask: "How does this cause a production outage?" or "How does this affect scalability?"
- **Observability:** Always link the theory to a /proc file or a CLI tool (e.g., strace, ss, lsof).

Week	Day	Task & TLPI Chapters	Interview Focus
WEEK 1: Process Lifecycle & Resource Isolation	Day 1	Process Memory Layout: Study TLPI Ch. 6. Difference between Stack, Heap, RSS, and VSZ.	Memory Leaks / OOM Analysis
	Day 2	Process Creation: Study fork(), exec(), and Copy-on-Write (CoW) .	Fork bombs / Resource exhaustion
	Day 3	Zombies & Orphans: Handling process termination and wait().	Cleanup / Init systems (tini)
	Day 4	Resource Limits (rlimits): Study TLPI Ch. 36. ulimit and setrlimit.	"Too many open files" errors
	Day 5	Scheduling Priorities: Study TLPI Ch. 35. Nice values and ionice.	Noisy Neighbors / CPU Throttling
	Day 6	Practical: Use strace to trace a process lifecycle.	System Call Overhead
	Day 7	Review: Scenario-based questions on OOM Killer behavior.	SRE Systems Thinking
WEEK 2: File Systems & Storage (Updated)	Day 1	File Descriptors & Inodes: Study TLPI	Storage Architecture

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		Ch. 4 & 14. VFS Layer.	
	Day 2	I/O Buffering: Study TLPI Ch. 13. Kernel vs. User space buffering (Stdio).	Disk Latency / Throughput
	Day 3	The Block Layer: Understanding /dev/sdX, LUN IDs, and HBA paths.	Infrastructure Storage Internals
	Day 4	Scanning & Resizing: Scanning SCSI buses via /sys/class/scsi_host and resizing FS live.	Online Storage Expansion
	Day 5	Multipathing: multipath-tools and handling redundant paths to SAN storage.	High Availability & Failover
	Day 6	Practical: Identify disk-bound processes using iostat and lsof.	Bottleneck Detection
	Day 7	Review: Explain what happens when a file is deleted but the space isn't freed.	Root Cause Analysis (RCA)
WEEK 3: Networking & Sockets (The SRE Core)	Day 1	The TCP Handshake/Teardown: TLPI Ch. 58. Every state (SYN_RECV, etc).	Connection Failures
	Day 2	TIME_WAIT & Port Exhaustion: Study TLPI Ch. 61. SO_REUSEADDR.	High-scale Networking
	Day 3	The Socket Buffer: rmem/wmem tuning for high latency links.	Throughput Optimization
	Day 4	Event Loops (epoll): Study TLPI Ch. 63. Level vs. Edge Triggering.	Load Balancer Internals
	Day 5	DNS & Name Resolution: /etc/resolv.conf and nsswitch.conf internals.	Service Discovery
	Day 6	Practical: Use ss and	Traffic Analysis

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		tcpdump to debug a connection reset (RST).	
	Day 7	Review: Packet journey from User-space to NIC.	Deep Stack Knowledge
WEEK 4: Signals, IPC & Advanced Isolation	Day 1	Signals: Study TLPI Ch. 20. SIGTERM vs SIGKILL vs SIGSTOP.	Graceful Shutdowns
	Day 2	Inter-Process Communication: Pipes and Shared Memory (shm).	Container Communication
	Day 3	Capabilities: Study TLPI Ch. 39. Running without Root.	Infrastructure Security
	Day 4	Uninterruptible Sleep (D State): Causes and impacts.	Kernel Lockups
	Day 5	Context Switching: Costs of switching between Kernel and User mode.	CPU Performance
	Day 6	Mock Interview: Troubleshooting a "High Load Average, Low CPU" scenario.	Senior Readiness
	Day 7	Final Review: Mapping TLPI concepts to Kubernetes resource limits/requests.	Platform Engineering

WEEKLY OUTPUT FORMAT

Each week, we will follow this loop:

1. **Reading:** You read the specified TLPI chapters.
2. **Observation:** You find the real-world metric for that concept on a Linux machine.
3. **Scenario:** I provide a production incident ("The site is down...") and you solve it using the week's theory.

Document created for Babu S — Focus: Infrastructure Engineering